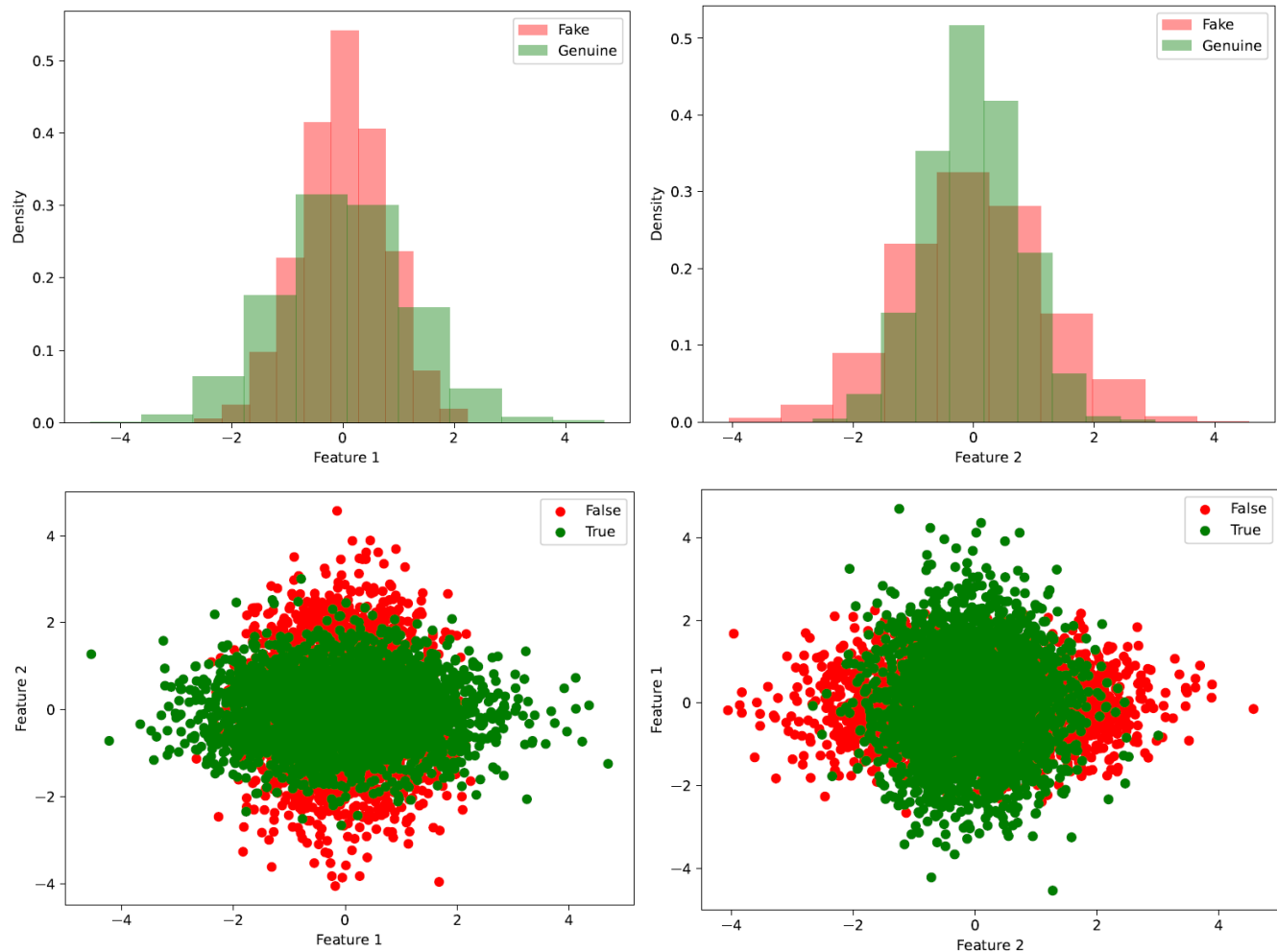


Lab. 2 - Project Report

1. Analyze the first two features. What do you observe? Do the classes overlap? If so, where? Do the classes show similar mean for the first two features? Are the variances similar for the two classes? How many modes are evident from the histograms (i.e., how many “peaks” can be observed)?



From the scatter plot, we observe that the Fake and Genuine classes have significant overlap, particularly in the middle region of the feature space.

From the histograms, the peaks of the distributions for Fake and Genuine fingerprints seem to be close. This indicates that the means of the first two features for both classes are relatively similar.

- The means for the Fake class are 0.00287744 (Feature 1) and 0.01869317 (Feature 2).
- The means for the Genuine class are 5.4454483e-04 (Feature 1) and -8.5243722e-03 (Feature 2).

The spread of the distributions in both histograms suggests that the variance for Fake and Genuine classes might be different. A deeper statistical analysis would confirm whether one class has a significantly higher spread than the other.

- The variance for the Fake class is 0.56958044 (Feature 1) and 1.4208633 (Feature 2).
- The variance for the Genuine class is 1.4302332 (Feature 1) and 0.57827723 (Feature 2).

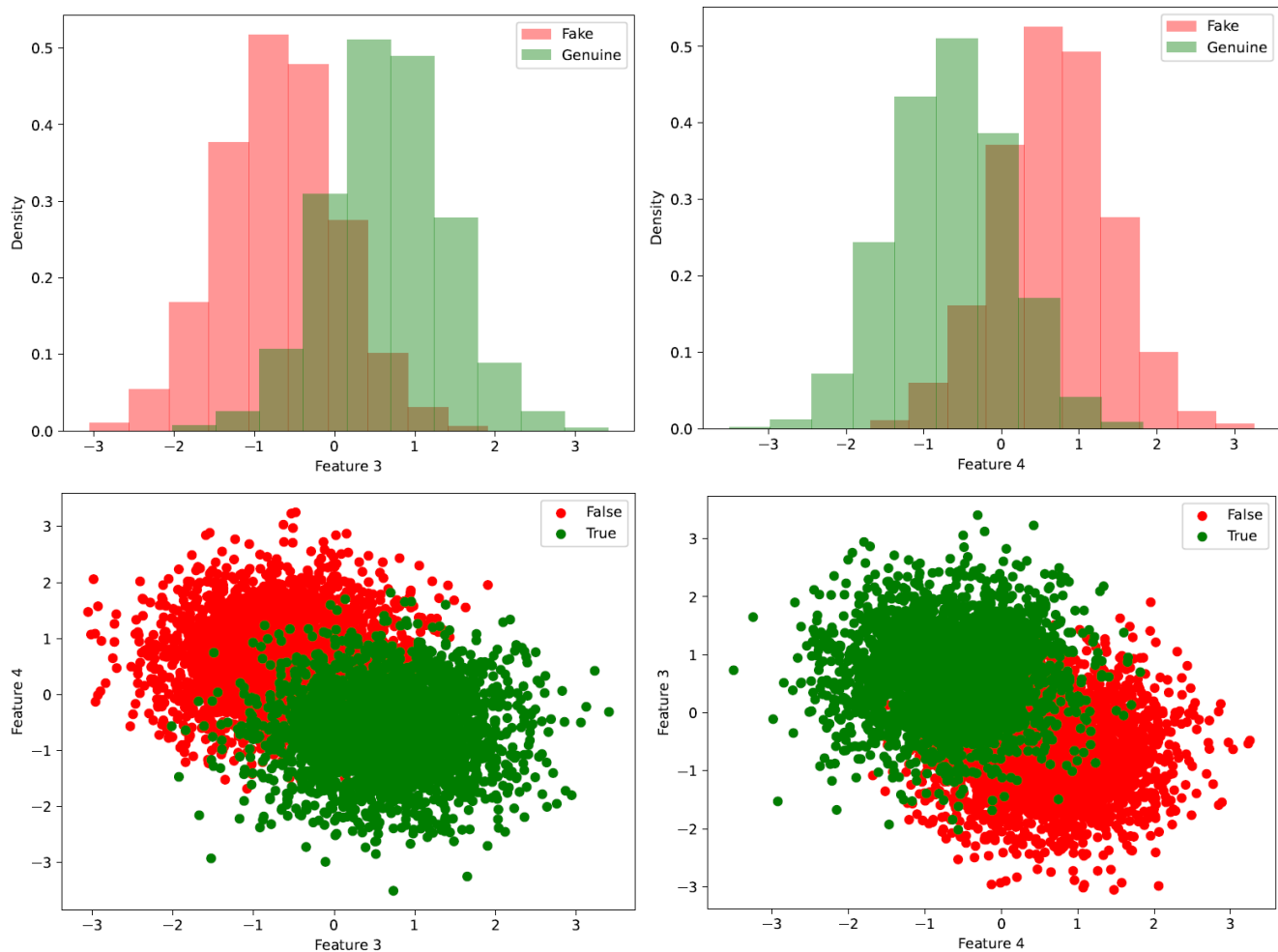
Both histograms seem to exhibit one main peak per class, suggesting a uni-modal distribution. This implies that each class is relatively well-defined but still overlaps with the other.

Conclusion

- Significant overlap.

- Similar means but different variances between classes.
- Uni-modal distribution per class.

2. Analyze the third and fourth features. What do you observe? Do the classes overlap? If so, where? Do the classes show similar mean for these two features? Are the variances similar for the two classes? How many modes are evident from the histograms?



Based on the data provided, the third and fourth features seem to show class separation, but some overlap still exists.

- The means for the Fake class are -0.68093944 (Feature 3) and 0.6708359 (Feature 4).
- The means for the Genuine class are 0.66523772 (Feature 3) and -0.6641956 (Feature 4).

The values are nearly symmetrical but opposite in sign, suggesting that these features might be useful for classification.

- The variance for Fake class is 0.54997754 (Feature 3) and 0.53604287 (Feature 4).
- The variance for Genuine class is 0.5489022 (Feature 3) and 0.5533431 (Feature 4).

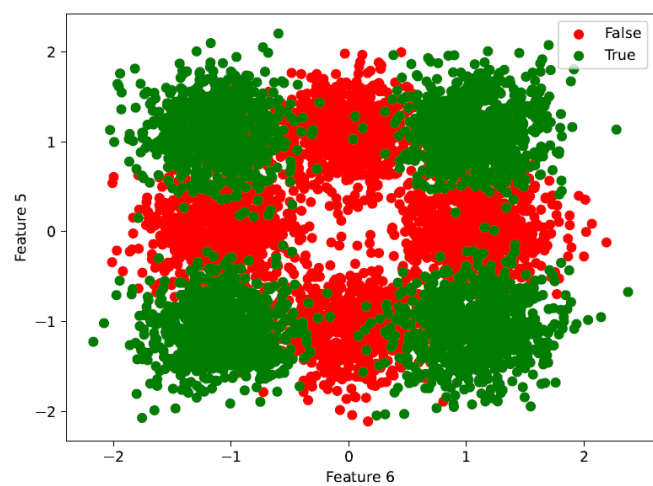
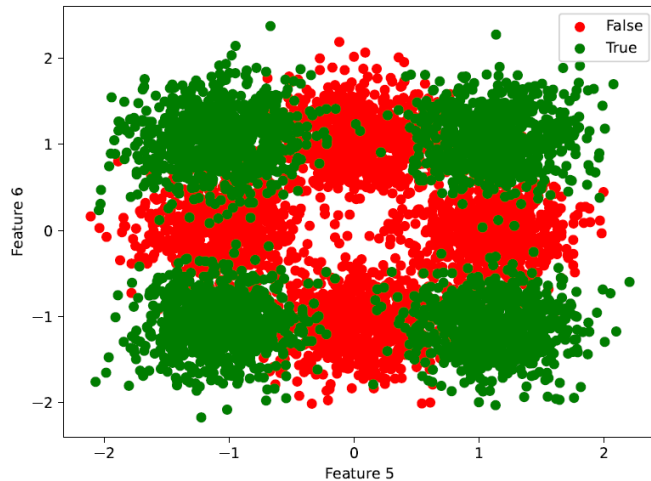
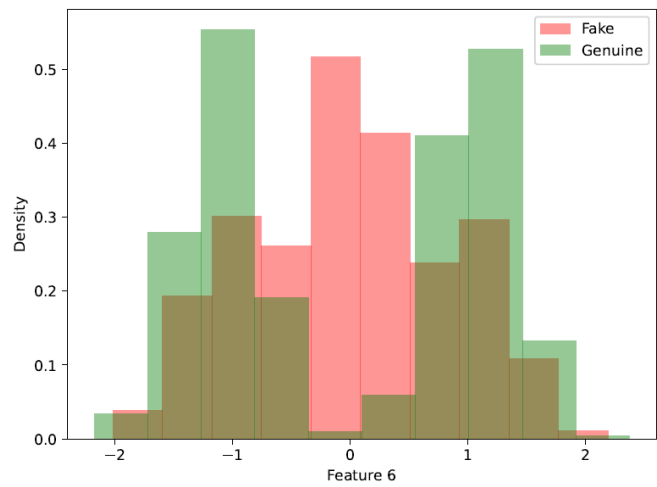
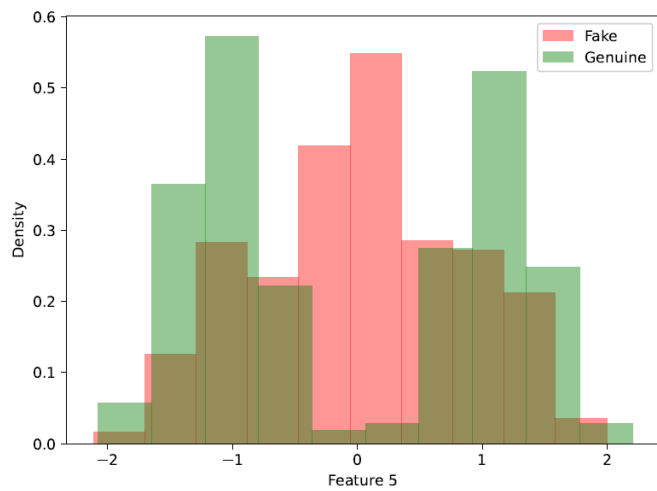
These values are quite similar, indicating that both classes exhibit comparable spreads in these two features.

The histograms likely display one peak per class, suggesting a bi-modal distribution for both classes.

Conclusion

- Better class separation with opposite sign means.
- Similar variances between classes.
- Bi-modal distributions.

3. Analyze the last two features. What do you observe? Do the classes overlap? If so, where? How many modes are evident from the histograms? How many clusters can you notice from the scatter plots for each class?



These features likely show significant overlap, but specific scatter plot information is needed for confirmation.

Since each class has a distinct mean but possibly overlapping distributions, the histograms should exhibit one peak per class.

Given that the variance values differ significantly between the two classes:

- Fake class variance: 0.6800733 (Feature 5) and 0.70503753 (Feature 6).
- Genuine class variance: 1.3177668 (Feature 5) and 1.2870293 (Feature 6).

The Genuine class has higher variance, meaning its points are more dispersed.

The Fake class has a tighter cluster distribution.

If we analyze clustering visually:

- Fake class (0): Likely three cluster.
- Genuine class (1): Possibly four clusters due to greater variance.

Conclusion

- Overlap exists, but the variance differs significantly.
- Fake class appears more compact, while Genuine class is more dispersed.
- Multi-modal distribution.

Mean, covariance matrix, variance, and standard deviation for the entire dataset.

- *Mean:*
0.00170711 0.00503903 -0.00560752 0.00109537 -0.00700025 0.00910515
- *Covariance matrix:*
[[1.0013431e+00 5.1675579e-03 1.7674640e-03 1.4526877e-02 1.0006677e-02 -1.2959122e-02]
[5.1675579e-03 9.9835277e-01 -1.5835416e-02 -4.4064570e-04 -4.0130778e-03 -8.1768673e-04]
[1.7674640e-03 -1.5835416e-02 1.0024818e+00 -4.3706036e-01 -2.6860058e-02 6.3438802e-03]
[1.4526877e-02 -4.4064570e-04 -4.3706036e-01 9.9029392e-01 2.5335757e-02 8.2692970e-03]
[1.0006677e-02 -4.0130778e-03 -2.6860058e-02 2.5335757e-02 1.0011975e+00 8.7578996e-04]
[-1.2959122e-02 -8.1768673e-04 6.3438802e-03 8.2692970e-03 8.7578996e-04 9.9722379e-01]]
- *Variance:*
1.0013431 0.9983527 1.0024818 0.99029386 1.0011975 0.9972238
- *Standard deviation:*
[1.0006714 0.999176 1.0012401 0.99513507 1.0005985 0.9986109]

Mean, covariance matrix, variance, and standard deviation for each class.

Class 0

- *Mean:*
0.00287744 0.01869317 -0.68093944 0.6708359 0.02795693 -0.00582741
- *Covariance matrix:*
[[5.6958103e-01 4.4340142e-03 8.4317056e-03 1.4447930e-02 1.4267275e-02 -1.4115002e-03]
[4.4340142e-03 1.4208657e+00 -4.7580758e-03 -1.2243296e-02 -2.5728114e-03 -5.5409307e-03]
[8.4317056e-03 -4.7580758e-03 5.4997694e-01 1.3203588e-02 -1.1525188e-02 1.1104402e-02]
[1.4447930e-02 -1.2243296e-02 1.3203588e-02 5.3604269e-01 7.0738117e-03 1.5974429e-02]
[1.4267275e-02 -2.5728114e-03 -1.1525188e-02 7.0738117e-03 6.8007350e-01 1.2000162e-02]
[-1.4115002e-03 -5.5409307e-03 1.1104402e-02 1.5974429e-02 1.2000162e-02 7.0503855e-01]]
- *Variance:*
0.56958044 1.4208633 0.54997754 0.53604287 0.6800733 0.70503753
- *Standard deviation:*
[0.75470555 1.1919997 0.7416047 0.7321495 0.8246656 0.8396651]

Class 1

- *Mean:*
5.4454483e-04 -8.5243722e-03 6.6523772e-01 -6.6419560e-01 -4.1725226e-02 2.3938498e-02
- *Covariance matrix:*
[[1.4302334 0.0058646 -0.00328748 0.01305324 0.00569338 -0.02439541]
[0.0058646 0.57827795 -0.0085804 -0.006824 -0.00638891 0.00427788]
[-0.00328748 -0.0085804 0.54890263 0.01126717 0.00465291 -0.01835332]
[0.01305324 -0.006824 0.01126717 0.5533427 -0.00288255 0.02041833]
[0.00569338 -0.00638891 0.00465291 -0.00288255 1.3177681 -0.00914105]
[-0.02439541 0.00427788 -0.01835332 0.02041833 -0.00914105 1.2870262]]
- *Variance:*
1.4302332 0.57827723 0.5489022 0.5533431 1.3177668 1.2870293
- *Standard deviation:*
[1.1959236 0.7604454 0.74087936 0.7438704 1.1479403 1.1344731]